

DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands.

Description:

PRIMARY KEY:

The PRIMARY KEY constraint uniquely identifies each record in a database table.

Primary keys must contain UNIQUE values, and cannot contain NULL values.

A table can have only one primary key, which may consist of single or multiple fields.

FOREIGN KEY:

- A FOREIGN KEY is a key used to link two tables together.
- A FOREIGN KEY is a field (or collection of fields) in one table that refers to the PRIMARY KEY in another table.
- The table containing the foreign key is called the child table, and the table containing the candidate key is called the referenced or parent table.

UNIQUE Constraint:

- The UNIQUE constraint ensures that all values in a column are different.
- Both the UNIQUE and PRIMARY KEY constraints provide a guarantee for uniqueness for a column or set of columns.
- A PRIMARY KEY constraint automatically has a UNIQUE constraint.
- However, you can have many UNIQUE constraints per table, but only one PRIMARY KEY constraint per table.

CHECK Constraint:

- The CHECK constraint is used to limit the value range that can be placed in a column
- If you define a CHECK constraint on a single column it allows only certain values for this column.
- If you define a CHECK constraint on a table it can limit the values in certain columns based on values in other columns in the row.

6)

PRIMARY:

```
ALTER TABLE table_name  
ADD PRIMARY KEY(primary_key_column);
```

FOREIGN KEY:

```
ALTER TABLE table_name  
ADD CONSTRAINT constraint_name  
FOREIGN KEY foreign_key_name (columns)  
REFERENCES parent_table(columns)  
ON DELETE action  
ON UPDATE action
```

UNIQUE:

```
CREATE TABLE table_1(  
...  
column_name_1 data_type,  
...  
UNIQUE(column_name_1)  
);
```

CHECK

```
CREATE TABLE IF NOT EXISTS parts (  
part_no VARCHAR(18) PRIMARY KEY,  
description VARCHAR(40),  
cost DECIMAL(10 , 2 ) NOT NULL CHECK(cost > 0), price DECIMAL (10,2) NOT  
NULL  
);
```

6)

Questions:

- 1) Alter the table STUDENT2 with following structure.

#	Column Name	Constraints
1	RegNo	PRIMARY KEY

- 2) Alter the table name FACULTY with following structure.

#	Column Name	Constraints
1	FacNo	PRIMARY KEY
2	Gender	CHECK 'M' or 'F'

- 3) Alter the table name DEPARTMENT with following structure.

#	Column Name	Constraints
1	DeptNo	PRIMARY KEY

- 4) Alter the table name COURSE with following structure.

#	Column Name	Constraints
1	CourseNo	PRIMARY KEY
2	SemNo	1 to 6

- 5) Alter the table name STUDENT2 with following structure.

#	Column Name	Constraints
1	S1Name	UNIQUE KEY

- 6) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table.

6)

OUTPUTS:

1)

```
mysql> DESC STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int           | NO   | PRI | NULL    |       |
| S1Name | varchar(30)   | YES  | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.066 sec)
```

2)

```
mysql> DESC FACULTY;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | int           | NO   | PRI | NULL    |       |
| FacName | varchar(30)   | YES  |     | NULL    |       |
| Gender | char(1)       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.029 sec)
```

6)

3)

```
mysql> DESC DEPARTMENT;
```

Field	Type	Null	Key	Default	Extra
DeptNo	int	NO	PRI	NULL	
DeptName	varchar(30)	YES		NULL	

2 rows in set (0.024 sec)

4)

```
mysql> DESC COURSE;
```

Field	Type	Null	Key	Default	Extra
CourseNo	int	NO	PRI	NULL	
CourseName	varchar(30)	YES		NULL	
SemNo	int	YES		NULL	

3 rows in set (0.028 sec)

5)

```
mysql> DESC ENROLLMENT;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	YES	MUL	NULL	
CourseNo	int	YES		NULL	

2 rows in set (0.021 sec)

6)

```
mysql> desc STUDENT2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(30)	YES	UNI	NULL	

2 rows in set (0.024 sec)

```
mysql> desc STUDENT3;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	YES		NULL	
S2Name	varchar(30)	YES		NULL	

2 rows in set (0.025 sec)

6)

Result:

DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.